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# Solutions to JEE Main Home Practice Test - 2 | JEE - 2024

**PHYSICS**

**SECTION-1**

**1.(B)** The electric filed at the centre of a charged semicircular wire is given by $E = 2\left[\frac{1}{4\pi\varepsilon_0}\frac{Q}{\pi r^2}\right]$ ($r$ = radius of semicircular wire)

$$\Rightarrow \quad E_{net} = 2\left[\frac{1}{4\pi\varepsilon_0}\right]\left[\frac{Q}{\pi r^2} - \frac{Q}{\pi R^2}\right] = \frac{2Q}{4\pi\varepsilon_0\pi}\left[\frac{1}{r^2} - \frac{1}{R^2}\right]$$

$$= \frac{2\times 0.7\times 10^{-9}\times 9\times 10^9}{3.14}\left[\frac{1}{(0.1)^2} - \frac{1}{(0.2)^2}\right] = \frac{2\times 0.7\times 9}{3.14}\left[10^2 - \frac{10^2}{4}\right]$$

$$= \frac{2\times 0.7\times 9\times 10^2(0.75)}{3.14} = 3.009\times 10^2 \approx 301 V/m$$

**2.(C)** $T_1 = 2\pi\sqrt{\frac{l}{g}}$ or $\frac{4\pi^2 l}{T_1^2} = g$

$T_2 = 2\pi\sqrt{\frac{l}{g+a}}$ or $\frac{4\pi^2 l}{T_2^2} = g + a$

$T_3 = 2\pi\sqrt{\frac{l}{g-a}}$ or $\frac{4\pi^2 l}{T_3^2} = g - a$

$\therefore \quad \frac{4\pi^2 l}{T_2^2} + \frac{4\pi^2 l}{T_3^2} = 2g$ where $g = \frac{4\pi^2 l}{T_1^2}$

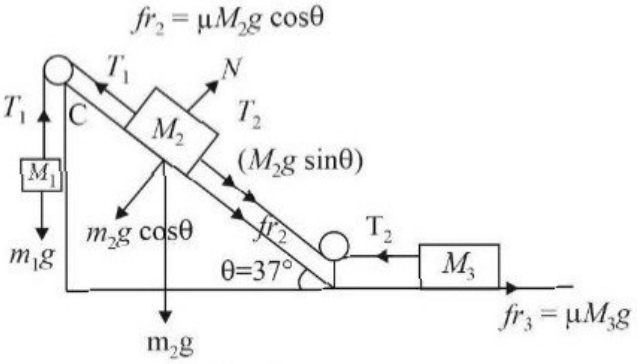
Solving we get : $T_1 = \frac{\sqrt{2}T_2T_3}{\sqrt{T_2^2 + T_3^2}}$

**3.(C)** Draw FBD for all the three blocks and apply Newton' second law for all the three blocks. Then, solve the expression to get answer.

$f_r =$ frictional force

For block $M_1$, $m_1 g = T_1$ ..... (i)

For block $M_2$, [As $fr_2 = \mu m_2 g\cos\theta$]




$T_1 = T_2 + M_2 g\sin(37^\circ) + \mu M_2 g\cos 37^\circ$

$M_1 g = T_2 + M_2 g(\sin 37^\circ + \mu \cos 37^\circ)$ ..... (ii)

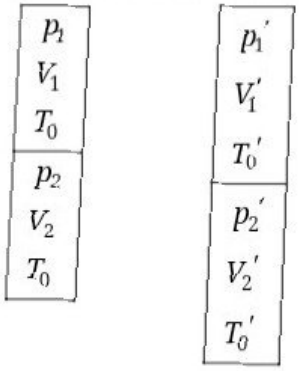
For lock $M_3$, $T_2 = \mu M_3 g$ $[f_3 = \mu M_3 g]$ ..... (iii)

From Eqs. (ii) and (iii), we get

$\Rightarrow \quad M_1 g = \mu M_3 g + M_2 g(\sin 37^\circ + \mu \cos 37^\circ)$

$\Rightarrow \quad M_1 = 0.25 \times 4 + 4\left(\frac{3}{5} + 0.25 \times \frac{4}{5}\right) = 1 + \frac{16}{5} = \frac{21}{5} = 4.2kg$

**4.(D)** The situation is shown in figure.




If m be the mass of piston and s its cross-sectional area, then $p_1 + \frac{mg}{s} = p_2$ and $p_1' + \frac{mg}{s} = p_2'$

From these equations, we have $p_2 - p_1 = p_2' - p_1'$ ..... (i)

For an ideal gas $p_1 = \frac{RT_0}{V_1}$, $p_2 = \frac{RT_0}{V_2}$

$p_1' = \frac{RT}{V_1'}$, $p_2' = \frac{RT}{V_2'}$

Substituting these values in Eq. (i), we get

$RT_0\left(\frac{1}{V_2} - \frac{1}{V_1}\right) = RT\left(\frac{1}{V_2'} - \frac{1}{V_1'}\right)$ or $\frac{T_0}{V_1}\left(\frac{V_1}{V_2} - 1\right) = \frac{T}{V_1'}\left(\frac{V_1'}{V_2'} - 1\right)$

or $T = T_0\left(\frac{\eta - 1}{\eta' - 1}\right)\frac{V_1'}{V_1}$ ..... (ii)

where $\frac{V_1}{V_2} = \eta$ and $\frac{V_1'}{V_2'} = \eta'$

From figure, $V_1' + V_2' = V_1 + V_2$ or $V_1'\left[1 + \frac{V_2'}{V_1'}\right] = V_1\left[1 + \frac{V_2}{V_1}\right]$

or $\frac{V_1'}{V_1} = \frac{\left[1 + (1/\eta)\right]}{\left[1 + (1/\eta')\right]} = \frac{\eta'(\eta + 1)}{\eta(\eta' + 1)}$ ..... (iii)

From Eqs. (ii) and (iii), we get $T = T_0\left[\left(\frac{\eta - 1}{\eta' - 1}\right)\left\{\frac{\eta'(\eta + 1)}{\eta(\eta' + 1)}\right\}\right]$

Here, $T_0 = 300K$, $\eta = 4$ and $\eta' = 3$

Substituting these values in Eqs. we get

$T = 300\left[\left(\frac{4 - 1}{3 - 1}\right)\left\{\frac{3(4 + 1)}{4(3 + 1)}\right\}\right] = 300\left[\frac{3}{2}\left\{\frac{3}{4} \times \frac{5}{4}\right\}\right] = 421.9K$

**5.(A)** $F_r = -\frac{du}{dr} = -Kr$

For circular Motion

$|F_r| = K_r = \frac{mV^2}{r}$

$\Rightarrow \quad kr^2 = mV^2 \qquad \ldots.(i)$

Bohr's Quantization $mvr = \frac{nh}{2\pi} \qquad \ldots..(ii)$

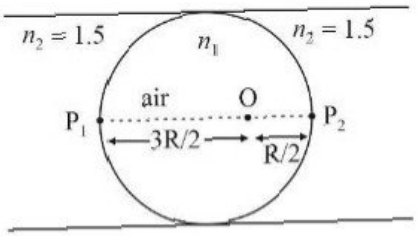
From (i) and (ii) $\frac{m^2V^2}{m} = kr^2$

$\Rightarrow \quad r = \left(\frac{h^2}{4\pi^2 mk}\right)^{1/4} n^{1/2} \Rightarrow r \propto \sqrt{n}$

$K.E. = \frac{1}{2}mV^2 \; PE = \frac{1}{2}kr^2 + \frac{kr^2}{2} = kr^2 \propto n$

**6.(C)** For the refraction through curved surface $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$

Refraction through left surface $\frac{3/2}{v_1} - \frac{1}{\left(\frac{-3}{2}R\right)} = \frac{(3/2-1)}{-R}$




$\frac{3}{2v_1} = \frac{(-3-4)}{6R}$

$\frac{3}{2v_1} = \frac{-7}{6R}$

$v_1 = \frac{-9}{7}R = -1.28R \quad \Rightarrow \quad 1.28\,R$ from $p_1$

Refraction through right surface

$\frac{3/2}{v_2} - \frac{1}{(-R/2)} = \frac{(3/2-1)}{-R}$

$\frac{3}{2v_2} + \frac{2}{R} = \frac{-1}{2R}$

$\frac{3}{2v_2} = \frac{-1}{2R} - \frac{2}{R}$

$\frac{3}{2v_2} = \frac{(-1-4)}{2R}$

$\frac{3}{2v_2} = \frac{-5}{2R}$

$v_2 = \frac{-3R}{5} = -0.6R \quad \Rightarrow$ So separation between the two images

$= R-(0.28+0.6)R = R-[0.88]R = 0.12R$

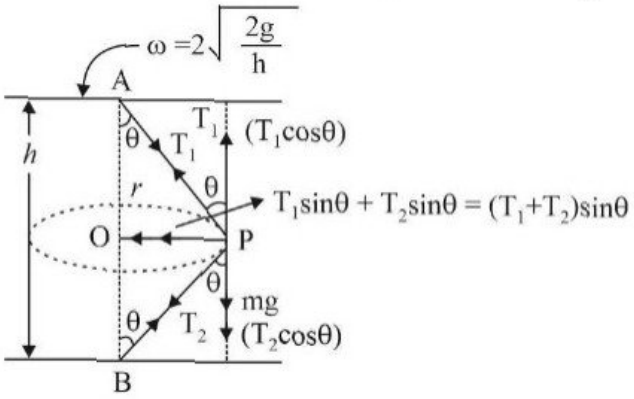
**7.(B)** From the figure it is clear that object P is tied with identical strings, so $\angle OAP = \angle PBO = \theta$ Here, $(T_1+T_2)\sin\theta$ will provide required centripetal force.

So, $(T_1+T_2)\sin\theta = mr\omega^2$ ..... (i)

and $T_1\cos\theta = mg + T_2\cos\theta \quad \cos\theta(T_1-T_2) = mg$ ..... (ii)

By Eqs. (i) and (ii), we get $\frac{(T_1+T_2)\sin\theta}{(T_1-T_2)\cos\theta} = \frac{r\omega^2}{g}$

$\frac{(T_1+T_2)}{(T_1-T_2)}\tan\theta = \frac{\omega^2}{g}\left[\tan\theta\times\frac{h}{2}\right]\left[\because \tan\theta \approx \frac{r}{(h/2)}\right]$




$\frac{(T_1+T_2)}{(T_1-T_2)} = \frac{\omega^2 h}{2g} = \left(4\times\frac{2g}{h}\right)\times\frac{h}{2g} \quad \left(\text{As, } \omega = 2\sqrt{2g/h}\right)$

$(T_1+T_2) = 4(T_1-T_2) \quad \Rightarrow \quad T_1+T_2 = 4T_1-4T_2$

$3T_1 = 5T_2 \quad \Rightarrow \quad \frac{T_1}{T_2} = \frac{5}{3}$

**8.(C)** $\lambda_d = \frac{h}{mV}$ (De'Broglie wavelength of electron)

$\lambda_p = \frac{C}{\nu}$ (photon)

So, $\frac{h}{mV} = \frac{C}{\nu}\times 10^{-3}$ (Given condition)

Solving, $V = 1.46\times 10^6 m/s$

**9.(A)** de-Broglie wavelength $\lambda = \frac{h}{\sqrt{2mk}}\left(As\,\lambda = \frac{h}{p}\right)$

$0.5\times 10^{-9} = \frac{6.6\times 10^{-34}}{\sqrt{2\times m_{He}\times\frac{3}{2}KT}}$ [k = kinetic energy]

$0.5\times 10^{-9} = \frac{6.6\times 10^{-34}}{\sqrt{4\times 1.67\times 10^{-27}\times 3\times 1.38\times 10^{-23}\times T}}$

Solving $T \approx 6.6K$

**10.(B)** Magnetic field at the centre of orbit $B=\frac{\mu_0 I}{2r}$ ..... (i)

$$I=\frac{e}{T}=\frac{e}{\frac{2\pi r}{v}}=\frac{ev}{2\pi r} \quad \text{..... (ii)}$$

From Eqs. (i) and (ii), we get $B=\frac{\mu_0}{2r}\left[\frac{ev}{4\pi r}\right]=\frac{\mu_0 ev}{4\pi r^2}$

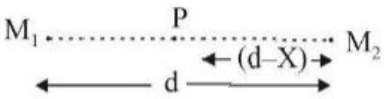
Dividing both sides by L

$$\Rightarrow \quad \frac{B}{L}=\frac{\frac{\mu_0}{4\pi}\frac{ev}{r^2}}{mvr}=\frac{\mu_0 e}{4\pi mr^3}(\text{As } L=mvr)$$

So, $B=\left(\frac{\mu_0 e}{4\pi mr^3}L\right)$

**11.(A)** Let at point P the net gravitational field is zero.

$$\Rightarrow \quad \frac{GM_1}{X^2}=\frac{GM_2}{(d-X)^2}$$




[$M_1$ = mass of 1st body, $M_2$ = mass of 2nd body, $G_1$ = gravitational constant]

$$\Rightarrow \quad \frac{M_1}{M_2}=\left(\frac{X}{d-X}\right)^2 \quad \Rightarrow \quad \frac{d-X}{X}=\sqrt{\frac{M_2}{M_1}} \quad \Rightarrow \quad \frac{d}{X}-1=\sqrt{\frac{M_2}{M_1}}$$

$$\Rightarrow \quad \frac{d}{X}=1+\sqrt{\frac{M_2}{M_1}} \quad \Rightarrow \quad \frac{d}{X}=\frac{\sqrt{M_1}+\sqrt{M_2}}{\sqrt{M_1}} \quad \Rightarrow \quad X=\frac{\left(\sqrt{M_1}\right)}{\sqrt{M_1}+\sqrt{M_2}}$$

At X net gravitational field is zero. So, net gravitational potential at P

[V = gravitational potential at point P]

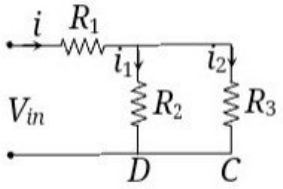
$$V=\frac{-GM_1}{X}+\left(\frac{-GM_2}{d-X}\right)=-G\left[\frac{M_1\left(\sqrt{M_1}+\sqrt{M_2}\right)}{d\sqrt{M_1}}+\frac{M_2\left(\sqrt{M_1}+\sqrt{M_2}\right)}{d\sqrt{M_2}}\right]$$

$$=\frac{-G}{d}\left[\left(\sqrt{M_1}+\sqrt{M_2}\right)\left(\sqrt{M_1}+\sqrt{M_2}\right)\right]=\frac{-G}{d}\left(\sqrt{M_1}+\sqrt{M_2}\right)^2$$

$$V=\frac{-G}{d}\left(M_1+M_2+2\sqrt{M_1M_2}\right)$$

**12.(C)** As the voltage in $R_2$ and $R_3$ is same therefore, according to,

$$H=\frac{V^2}{R}\cdot t, \quad R_2=R_3$$




Also the energy in all resistance is same.

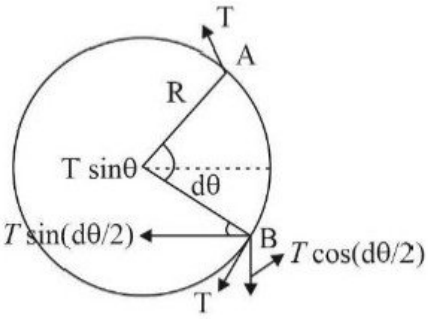
$$\therefore \quad i^2R_1t=i_1^2R_2t$$

Using $i_1=\frac{R_3}{R_2+R_3}i=\frac{R_3}{R_3+R_3}i=\frac{1}{2}i$

Thus, $i^2R_1t=\frac{i^2}{4}R_2t$ or $R_1=\frac{R_2}{4}$

**13.(A)** Let us consider a small Arc AB = dS

The components of T, $T\cos\left[\frac{(d\theta)}{2}\right]$ will cancel each other.




$2T\sin\left(\frac{d\theta}{2}\right)$ will provide the required centripetal force.

So, $2T\sin\left(\frac{d\theta}{2}\right)=(dm)R\omega^2$ (dm = mass of small arc)

For small angle, $\sin\left(\frac{d\theta}{2}\right)\approx\left(\frac{d\theta}{2}\right)$ [As if $\theta <<< \tan\theta=\sin\theta=\theta$]

$\Rightarrow \quad 2T\left(\frac{d\theta}{2}\right)=dmR\omega^2$ [$\rho$ = density, $\omega$ = angular velocity]

$Td\theta=dmR\omega^2$ ; $Td\theta=(dS\times\rho)R\omega^2\times A$

$T=\frac{dS}{d\theta}\times\rho R\omega^2\times A\left(R=\frac{dS}{d\theta}\right)$ ($dS$ : Arc length)

$T=\rho R^2\omega^2\times A$[$T$ = tension....(i)]

From Young's modules, $Y=\frac{T/A}{\Delta l/l}$ ..... (ii)

[$\Delta l$ = change in length]

Let the length of ring is $l$.

$\Rightarrow \quad l=2\pi r \Rightarrow \Delta l=2\pi\Delta R \quad \Rightarrow \quad \frac{\Delta l}{l}=\frac{\Delta R}{R}$ ..... (iii)

From Eqs. (ii) and (iii), we get $\frac{\Delta R}{R}=\frac{T}{AY}$ [$\Delta R$ = change in reading]

$\Rightarrow \quad \Delta R=\frac{RT}{AY}$ ..... (iv)

From Eqs. (i) and (iv), we get

$\Rightarrow \quad \Delta R=\frac{R}{AY}\left[\rho R^2\omega^2\times A\right]\Rightarrow\Delta R=\frac{\rho R^3\omega^2}{Y}$

**14.(B)** For the pitot tube by Bernaulli's theorem and Pascal's law $\frac{1}{2}\rho v^2=\Delta p=\rho_0 g\Delta h$

So, speed of the gas $v=\sqrt{\frac{2\rho_0 g\Delta h}{\rho}}$ ..... (i)

Now, rate of gas flow ($Q$), $Q=v\times s$ ..... (ii)

From (i) and (ii), we get $Q=\sqrt{\frac{2\rho_0 g\Delta h}{\rho}}\cdot s$

**15.(A)** $\frac{mgl}{2}+mgl=\frac{1}{2}\left(\frac{4}{3}ml^2\right)\omega^2$, $\omega=\frac{3}{2}\sqrt{\frac{g}{l}}$

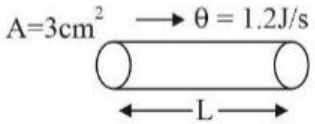
**16.(B)** A represents R and B represents to $X_L$.

So, $AB=RX_L=(\Omega)^2$ (unit wise)

$$=\left[\frac{V}{A}\right]^2=\left[\frac{W}{q\times A}\right]^2=\left(\frac{F\times S\times t}{q\times E}\right)$$

$$[AB]=\left(\frac{(MLT^{-1})\times(L)\times(T)}{[A]^2[T^2]}\right)^2=\left[ML^2T^{-3}A^{-2}\right]^2=\left[M^2L^4T^{-6}A^{-4}\right]$$

**17.(D)** If power P = 1.2W

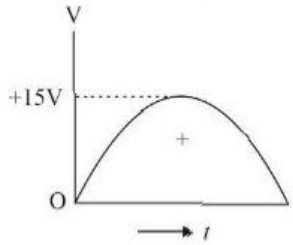



$K = \frac{400W}{mK}, \Delta T = 10°C = 10K$

Rote of heat flow $\frac{\Delta Q}{\Delta t} = \frac{KA\Delta T}{L}$

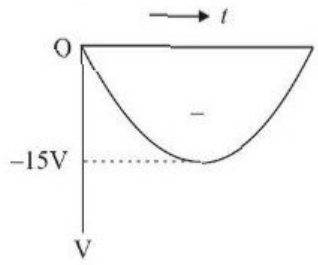
$1.2 = \frac{400 \times 3 \times 10^{-4} \times 10}{L} \Rightarrow L = \frac{12 \times 10^{3} \times 10^{-4} \times 10}{1.2} \Rightarrow L = 1m$

**18.(A)** For positive half,




$D_1$ will be forward and conduct current and $D_2$ will be revered biased and it will be not conduct.

To negative half.




$D_2$ will conduct and $D_1$ will not conduct.

**19.(C)** Given $\frac{di_1}{dt} = \frac{di_2}{dt}$

For an inductor $V = L\frac{di}{dt}$

$\therefore \quad \frac{V_2}{V_1} = \frac{I_2}{I_1} = \frac{1}{4}$

As instantaneous powers are equal $V_1 i_1 = V_2 i_2 \Rightarrow \frac{i_1}{i_2} = \frac{V_2}{V_1} = \frac{1}{4}$

$\therefore \quad \frac{W_1}{W_2} = \frac{L_1 i_1^2}{L_2 i_2^2} = (4)\left(\frac{1}{4}\right)^2 = \frac{1}{4} \Rightarrow \frac{W_2}{W_1} = 4$

**20.(D)** $\vec{R}_1 = \vec{F}_1 + \vec{F}_2, \ \vec{R}_2 = \vec{F}_1 - \vec{F}_2$

If $\alpha$ be the angle between $\vec{R}_1$ and $\vec{R}_2$ then $\cos\alpha = \frac{\vec{R}_1 . \vec{R}_2}{R_1 R_2} = \frac{F_1^2 - F_2^2}{F_1^2 + F_2^2}$ but $F_1 = F_2$

$\Rightarrow \quad \cos\alpha = 0$ and $\alpha = 90°$

## SECTION - 2

**1.(24)** $8 = 0 + a\left(2 - \frac{1}{2}\right)$ ...(1)

$S_5 = 0 + a\left(5 - \frac{1}{2}\right)$ ...(2)

Dividing equation (2) by (1)

We get, $S_5 = 24\,\text{m}$

**2.(10)** $|\Delta\vec{V}| = |\vec{V}_F - \vec{V}_i| = \sqrt{10^2 + 10^2 - 2\times10\times10\cos60^\circ} = \sqrt{100}m/s = 10\,m/s$

**3.(8)** $V = 5 + 4x^2$ [potential along x-axis]

$E = \frac{-dV}{dx} = -8x$ [Electric field along x-axis]

$E(x = -1) = 8\frac{V}{m}$

So, force along x-axis given by

$\Rightarrow \quad F = qE \approx 1\times8 = +8N$

**4.(15)** Tension in the rope = ?

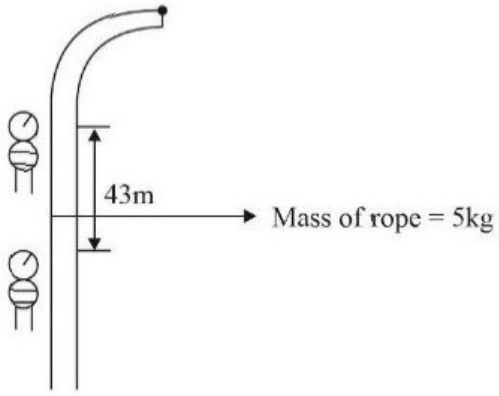
$\text{Speed} = \frac{\text{Distance}}{\text{Time}}, v = \frac{420m}{1.4s} = 300m/s$

Speed of wave on a string $v = \sqrt{T/\mu}$

[T = tension, μ = mass per unit length]

$300 = \sqrt{\frac{T}{\left(\frac{5}{420}\right)}} \quad \Rightarrow \quad (300\times300) = \frac{T\times420}{5}$

$T = \frac{300\times300\times5}{420} = \frac{15}{14}kN \quad \therefore \quad n = 15$




**5.(5)** Problem solving strategy Fist apply the condition for isothermal compression and then for adiabatic expansion. For isothermal compression,

$\Rightarrow \quad p_2 = 2p_1$

Isothermal $pV$ = constant

adiabatic $\rightarrow pV^\gamma$ = constant, $pV$ = constant

$\Rightarrow \quad p_1V_1 = p_2V_2, \; p_1V_1 = 2p_1V_2 \Rightarrow V_2/2$

For adiabatic expansion, $pV^\gamma$ = constant

$\Rightarrow \quad p_2V_2^\gamma = p_3V_1^\gamma \quad \Rightarrow \quad 2p_1\left(\frac{V_1}{2}\right)^\gamma = p_3(V_1)^\gamma \quad \left[\text{As } V_2 = \frac{V_1}{2}\right]$

$\Rightarrow \quad p_3 = 2p_1\left(\frac{1}{2}\right)^\gamma = 2p_1\left(\frac{1}{2}\right)^{7/5}$

$= 2^{1-\frac{7}{5}}p_1 = 2^{-2/5}p_1 \quad \Rightarrow \quad p_3 = \frac{p_1}{2^{2/5}} = 0.76p_1$

**6.(3)** $\vec{r}$ = position vector of particle $= 2\cos(\omega t)\hat{i} + 3\sin(\omega t)\hat{j}$

$$\vec{v} = \frac{d\vec{r}}{dt} = -2\omega\sin(\omega t)\hat{i} + 3\omega\cos(\omega t)\hat{j}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = -2\omega^2\cos(\omega t)\hat{i} - 3\omega^2\sin(\omega t)\hat{j} = -\omega^2\left[2\cos(\omega t)\hat{i} + 3\sin(\omega t)\hat{j}\right]$$

$$\vec{a} = -\omega^2\vec{r} \qquad \vec{F} = m\vec{a} = -m\omega^2\vec{r}$$

$$\Rightarrow \quad \text{Torque} = |\vec{r}\times\vec{F}| = |r||F|\sin\theta = r\times\left(-m\omega^2 r\right)\sin\pi\left[As\,\theta = 180^\circ\right] = \text{zero}$$

**7.(4)** $V_0 = 283V$

$$V_{rms} = \frac{V_0}{\sqrt{2}} = \frac{283}{2}\sqrt{2}V$$

$$X_L = \omega L = 320\times\frac{1}{40}\Omega = 8\Omega$$

$$X_C = \frac{1}{\omega C} = \frac{1}{320\times1000\times10^{-6}}\Omega = \frac{1000}{320}\Omega = \frac{25}{8}\Omega$$

$$X = X_L - X_C = 8 - \frac{25}{8} = \frac{39}{8}\Omega \approx 5\Omega$$

$$Z = \sqrt{R^2 + X^2} = 5\sqrt{2}\Omega = 5\times1.414\Omega = 7.\Omega$$

$$I_{rms} = \frac{V_{rms}}{Z} = \frac{283\sqrt{2}}{2\times5\sqrt{2}}A = 28.3A$$

$$\tan\phi = \frac{X}{R} = \frac{5}{5} = 1 \Rightarrow \phi = 45^\circ$$

**8.(40)** Heat needed to bring 50 gm water from $40^\circ C$ to $0^\circ C$

$$Q_1 = 50\times4.2\times40 = 8400J$$

Out of this some heat is provided by 20gm ice.

$$Q_2 = 20\times2.1\times20 = 840\text{ J}$$

$$Q = Q_1 - Q_2 = 7560$$

If mass '$m$' of ice is melted, then $7560 = m\times334 + m\times2.1\times20$ gives $m \cong 20\,gm$.

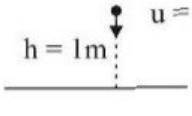
So, $\quad m(\text{total}) = 40gm$

**9.(90)** Coefficient of restitution $e = \dfrac{\sqrt{2gh'}}{\sqrt{2gh}}$

($h'$ = height rises when through with velocity $2m/s$, $h$ = height rises when dropped from $1m$)

Here, $\quad h' = \dfrac{3}{4}h$ and $h = 1m$

$$e = \sqrt{\frac{h'}{h}} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$$




Speed at the time touching the ground $v_1^2 = u^2 + 2gh = 4 + 2\times10\times1$

$v_1 = \sqrt{24}m/s$

So, $e = \dfrac{v_2}{v_1}$ $\quad [v_2$ = the speed which the ball will rise] $\dfrac{\sqrt{3}}{2} = \dfrac{v_2}{\sqrt{24}}$

$$v_2 = \frac{\sqrt{3\times24}}{2} = \sqrt{18}m/s$$

Height up to which the ball will rise $\quad v^2 = u^2 - 2gh \Rightarrow 0 = v_2^2 - 2gh'$

$v_2^2 = 2gh' \Rightarrow 18 = 2\times10\times h' \Rightarrow h' = 0.9\,m = 90\,cm$

**10.(25)** When corrective lens is used then eye can see the object at infinity. Power of eye lens in this situation is $p_\infty$

$u=\infty$ and $v=2cm=0.02m$

$$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$$

$$P_\infty=\frac{1}{0.02}-\frac{1}{\infty}=50$$

$$P_\infty=50+0$$

$$P_\infty=50D$$

If $P=$ Power of eye at near point when corrective lens is used

$$P=P_\infty+P_a=50+4=54D$$

Let near point in this situation is $x_n$

$$u=-x_n m$$

$$v=+2cm=0.02m$$

$$\frac{1}{f}=54$$

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{0.02}+\frac{1}{x_n}=54 \text{ (all distance are in m)}$$

$$50+\frac{1}{x_n}=54$$

$$x_n=\frac{1}{4}m=0.25m=25cm$$

## CHEMISTRY

### SECTION - 1

**1.(A)** Neptunium (Np) and plutonium (Pu) show maximum number of oxidation states starting from +3 to +7.

**2.(B)** $K_b = 10^{-5} \Rightarrow pK_b = 5 \Rightarrow \log 2 = 0.301$

$0.02M\ NH_4Cl$

$NH_4Cl + H_2O \rightleftharpoons NH_4OH + HCl$

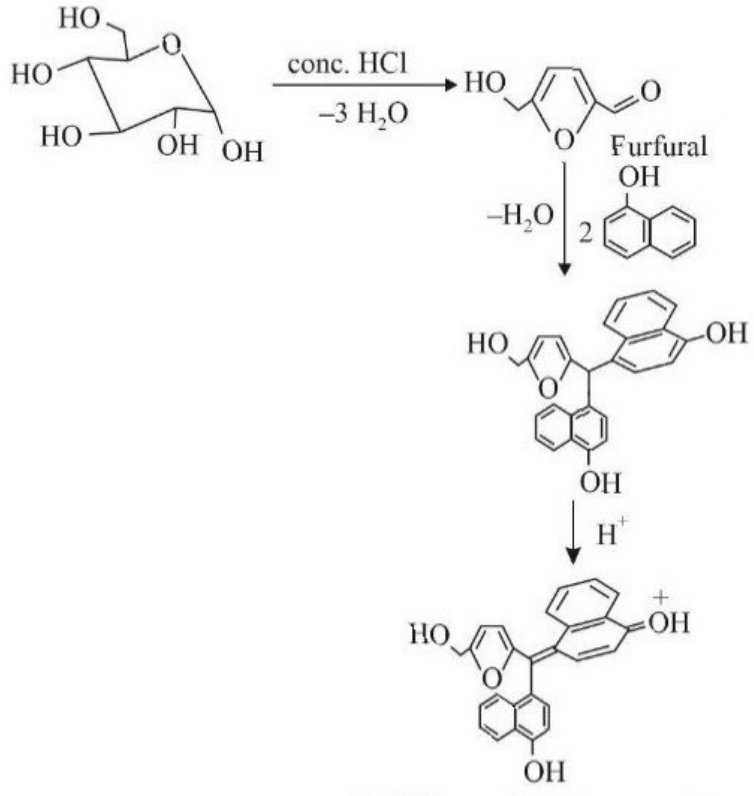
Acidic salt

$$pH = 7 - \frac{1}{2}(pK_b + \log C) = 7 - \frac{1}{2}\left(5 + \log 2\times 10^{-2}\right) = 7 - \frac{1}{2}(3.301) = 7 - 1.6505$$

$pH = 5.3495$

**3.(C)** $Cr_2O_3$ reacts with acid and alkali both where as $Sc_2O_3$ reacts with acid only.

**4.(A)** Molisch's test is a test for carbohydrates larger than tetroses. In molisch's test, the carbohydrate undergoes dehydration upon addition of conc. HCl or $H_2SO_4$. This molecule undergoes condensation with $\alpha$-naphthol present in reagent, resulting in the formation of purple or reddish-purple coloured complex




Reddish-purple colour complex.

**5.(C)** $K_1 = 2.5\times 10^{-4}\ dm^3\ mol^{-1}\ s^{-1}\ T_1 = 327^\circ C \Rightarrow 600K$

$K_2 = 1.0\ dm^3\ mol^{-1}\ s^{-1}\ T_2 = 527^\circ C \Rightarrow 800K$

$E_A \Rightarrow in\ kJ/mole$

Given $R = 8.314\ J/K\text{-}mole$

From $\log\frac{K_2}{K_1}=\frac{Ea}{2.303R}\left(\frac{1}{T_1}-\frac{1}{T_2}\right)$

$$\log\frac{1}{2.5}\times10^{-4}=\frac{Ea}{2.303R}\left(\frac{1}{T_1}-\frac{1}{T_2}\right)$$

$$\log\frac{1}{2.5\times10^{-4}}=\frac{Ea}{2.303\times8.314}\left(\frac{1}{600}-\frac{1}{800}\right)$$

$$E=165.54\,kJ$$

**6.(B)** Benzaldehyde can reduce Tollen's Reagent but not Fehling solution

**7.(B)** CFSE

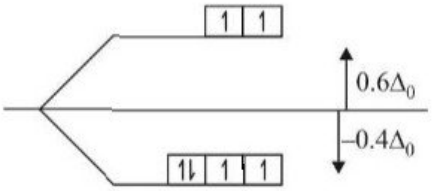
Of $\underset{\text{Octahedral Complex}}{\left[Fe(H_2O)_6\right]}Cl_2$

$H_2O$ weak field Ligand

i.e. do not pair up the unpaired electron.

$\left[Fe(H_2O)_6\right]^{+2}$

$Fe^{+2}\,1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^6$.



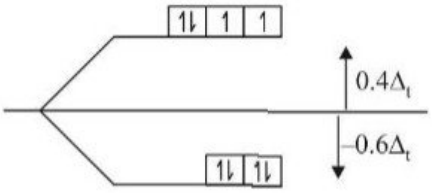

$CFSE \Rightarrow -4\times0.4\Delta_0+2\times0.6\Delta_0$

$=-1.6\Delta_0+1.2\Delta_0$

$=-0.4\Delta_0$

$K_2[NiCl_4]$

$\underset{Ni^{+2}}{[NiCl_4]^{-2}}$ Tetrahedral Complex $\underset{sp^3}{}$




$Cl^- \Rightarrow$ weak field Ligand do not pair up

$Ni^{+2}\quad 1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^8$

$CFSE=-0.6\times4\Delta_t+0.4\times4\Delta_t=-2.4\Delta_t+1.6\Delta_t$

$CFSE\approx-0.8\Delta_t$

**8.(C)** Given $2SO_2(g)+O_2(g)\rightleftharpoons 2SO_3(g)$

$\Delta H=-57.2\,kJ/mol$

$K_c\approx1.7\times10^{16}$

(A) From $K\downarrow=A\,e^{-\frac{\Delta H}{RT\uparrow}}$ as T increases, K decreases

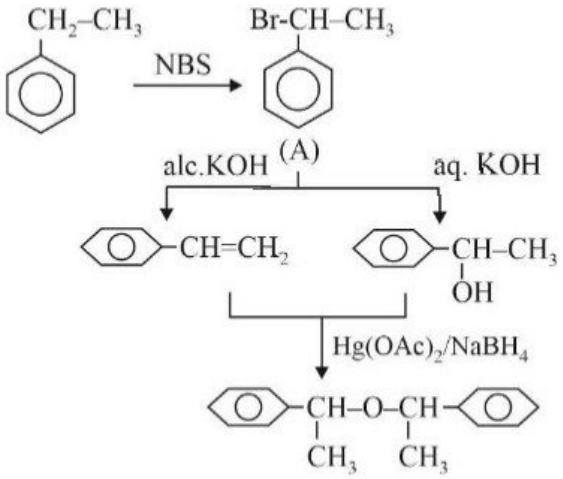
(B) Equilibrium constant is not a affected by change in volume.

(C) Although $K_c$ is large but it doesn't mean reaction go for completion.

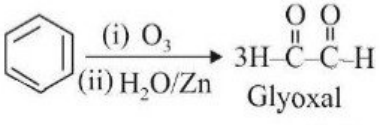
(D) Equilibrium will shift in forward direction as pressure increases.

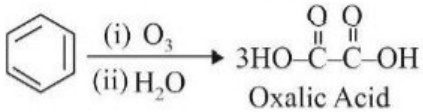
**9.(D)** In $SO_4^{2-}$ ion, sulphur can decrease its oxidation state only so it acts as only oxidising agent.

**10.(B)**



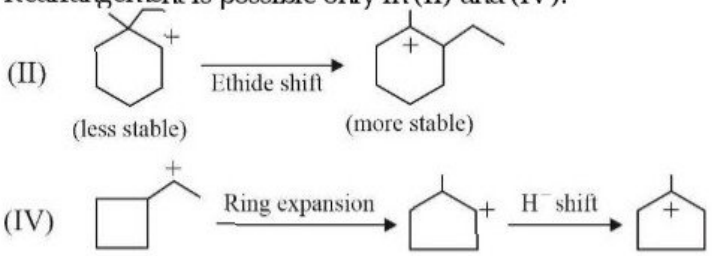

**11.(A)**




**12.(C)** Since, this compound have no carbon, $CN^-$ ion is not formed and presence of nitrogen can't be detected.

**13.(B)** Rearrangement is possible only in (II) and (IV).




**14.(A)** The wavelength of absorption for complex depending on value of $\Delta_0 = E = \frac{hc}{\lambda}$.

**15.(B)** **(A)** $sp^3$ **(B)** $dsp^2$ **(C)** $sp^3d^2$ **(D)** $sp^3d^2$

**16.(C)** All polysaccharides are non reducing sugars this is due to acetal glycosidic linkage present between monosaccharide units.

**17.(B)** $\underset{(X)}{2KMnO_4} \xrightarrow{\Delta} \underset{(Y)}{K_2MnO_4} + MnO_2 + \frac{1}{2}O_2$

$MnO_2 + 2Cl^- + 4H^+ \longrightarrow Mn^{+2} + 2H_2O + Cl_2$

**18.(B)** $Cd(OH)_2 \rightleftharpoons Cd^{+2} + 2OH^-$

$K_{sp} = [Cd^{+2}][OH^-]^2$

solubility $s = [Cd^{+2}] = \frac{K_{sp}}{[OH^-]^2}$ ;

pH of the buffer = 12 ; pOH = 2, $[OH] = 10^{-2}$ ; $K_{sp} = 4s^3 = 4(1.84\times10^{-5})^3$

$[Cd^{+2}]_f = \frac{4[1.84\times10^{-5}]^3}{(10^{-4})} = \frac{4\times6.23\times10^{-15}}{10^{-4}} = 2.49\times10^{-10}M.$

**19.(C)** $FeSO_4 \cdot 7H_2O$

(pale green salt)

$Fe^{2+} \xrightarrow{\text{standing in Air}} Fe^{3+}$

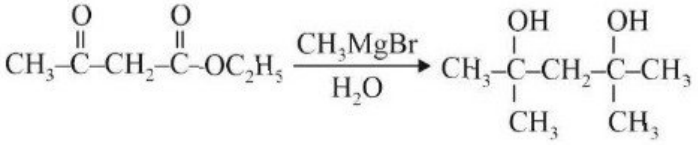
$Fe^{3+} + 3OH^- \longrightarrow \underset{\text{Brown}}{Fe(OH)_3}$

$Fe^{2+} + H_2S \xrightarrow{2OH^-} \underset{\text{Black ppt.}}{FeS} + 2H_2O$

$5Fe^{2+} + MnO_4^- + 8H^+ \longrightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$

**20.(B)** If $\Delta G^\circ < O$ then $K_{eql} > 1$

**SECTION - 2**

**1.(3)** 

**2.(750)**

| | NaOH | + HCl | ⟶ NaCl | + $H_2O$ |
|---|---|---|---|---|
| millimoles | = 250 × 0.5 | 500 × 1 | 0 | 0 |
| | = 125 | 500 | 125 | 125 |

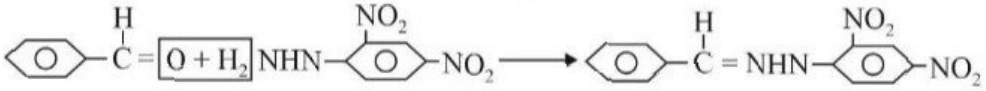
No. of millimoles of NaCl formed = 125

So, no. of moles = 0.125

No. of molecules = $0.125 \times 6 \times 10^{23} = 0.75 \times 10^{23} = 750 \times 10^{20}$

**3.(25)** Energy produced by eating 160 gm of glucose = $\frac{2800}{180} \times 160 = 2488.8$

Maximum distance travelled by person = $\frac{2488.8}{100} \approx 24.88\text{km} = 25\text{ km}$

**4.(4)** 

**5.(46)** $k = \frac{2.303}{60} \log \frac{100}{100-25}$

$k = \frac{2.303}{60} \log \frac{100}{75}$

$k = \frac{2.303}{60} (\log 4 - \log 3)$

$k = \frac{2.303}{60} (0.6 - 0.48)$

$k = 0.004606 = 46.06 \times 10^{-4}$

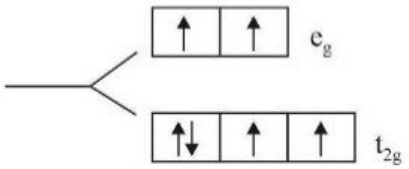
**6.(356)** $Y_{CCl_4} = \frac{X_{CCl_4} . P^0_{CCl_4}}{X_{CCl_4} . P^0_{CCl_4} + X_{SiCl_4} . P^0_{SiCl_4}}$

$$\frac{\frac{\frac{w}{154}}{w/154 + w/170} \times 115}{\frac{w/154}{w/154 + w/170} \times 115 + \frac{w/170}{w/154 + w/170} \times 230} \approx 0.356$$

**7.(6)** This include 2 double bond and 1 lone pair.

**8.(2)** $[CoF_6]^{3-}$ has $Co^{3+}$

$Co^{3+} = [Ar]3d^6$




**9.(18)** $(i-ii)\times 2 =$ equation (iii)

$$\therefore \quad k_p = \left(\frac{k_{p_1}}{k_{p_2}}\right)^2 = \left(\frac{1.8\times 10^8}{8\times 10^{-2}}\right)^2 = 0.05\times 10^{20} = 5\times 10^{18}$$

**10.(12)**

| S. No. | Electronic Configuration | $l$ | $m_l$ | $(l+m_l)$ | Electrons |
|---|---|---|---|---|---|
| 1 | $1s^2$ | 0 | 0 | 0 | 2 |
| 2 | $2s^2$ | 0 | 0 | 0 | 2 |
| 3 | $2p^6$ | 1 | –1 | 0 | 2 |
| | | 0 | | | |
| | | +1 | | | |
| 4 | $3s^2$ | 0 | 0 | 0 | 2 |
| 5 | $3p^6$ | 1 | –1 | 0 | 2 |
| | | 0 | | | |
| | | +1 | | | |
| 6 | $4s^2$ | 0 | 0 | 0 | 2 |

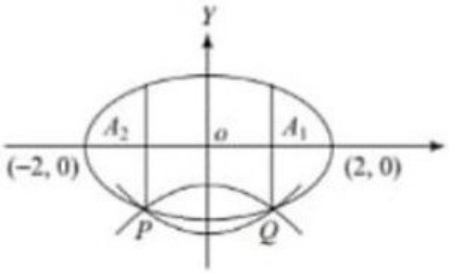
**MATHEMATICS**

**SECTION-1**

**1.(D)** Eccentricity $e$ of the ellipse is given by

$b^2 = a^2(1-e^2) \Rightarrow \quad 1 = 4\left(1-e^2\right) \Rightarrow e = \sqrt{3}/2.$

Focii of the ellipse are $\left(\sqrt{3},0\right)$ and $\left(-\sqrt{3},0\right)$




Length of a latus rectum of the ellipse is $2\dfrac{b^2}{a} = 1$

Thus, $P(x_1, y_1) = p\left(-\sqrt{3}, -1/2\right)$ and $Q(x_2, y_2) = Q\left(\sqrt{3}, -1/2\right)$

Length of the latus rectum $PQ$ of the parabola is $\quad |x_2 - x_1| = 2\sqrt{3} = 4p(say)$

As focus of a parabola is the mid point of the latus rectum, focus of the desired parabola is $(0, -1/2)$

and hence its vertices are $(0, -1/2 \pm p)$

i.e. $\left(0, -\dfrac{1}{2} - \dfrac{\sqrt{3}}{2}\right)$ and $\left(0, -\dfrac{1}{2} + \dfrac{\sqrt{3}}{2}\right)$

Thus there are two parabolas having $PQ$ as the latus rectum whose equations are

$x^2 = 4p\left(y + \dfrac{1}{2} + \dfrac{\sqrt{3}}{2}\right) = 2\sqrt{3}y + \sqrt{3} + 3$

And $x^2 = -4p\left(y + \dfrac{1}{2} - \dfrac{\sqrt{3}}{2}\right) = -2\sqrt{3}y - \sqrt{3} + 3$

**2.(B)** As $1 < 2x^2 - 3 < 2 \qquad \forall x \in \left(\sqrt{2}, \sqrt{\dfrac{5}{2}}\right)$

$\therefore \quad \displaystyle\int_{\sqrt{2}}^{\sqrt{5/2}} \left[2x^2 - 3\right] dx > 0 \qquad \forall x \in (0, 2)$

$\Rightarrow \quad g(x) = 0$ should have at least one real root in $(0, 2) \quad \{\because g'(x) \neq 0\}$

**3.(A)** Let remaining two observations are $x$ and $y$

$\therefore \quad x + y = 14$

Also $\quad x^2 + y^2 = 100 \quad \Rightarrow \quad |x - y| = 2$

**4.(D)** $\int_0^x f(t)dt = \int_x^1 t^2.f(t)dt + \frac{x^{16}}{8} + \frac{x^6}{3} + a$ ..... (i)

For x = 1, $\int_0^1 f(t)dt = 0 + \frac{1}{8} + \frac{1}{3} + a = \frac{11}{24} + a$

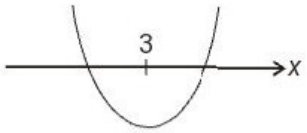
Diff. both sides of (i) w. r. t. x we get;

$f(x) = 0 - x^2 f(x) + 2x^{15} + 2x^5$

$\Rightarrow \quad 2\int_0^1 \frac{x^{15} + x^5}{1+x^2}dx = \frac{11}{24} + a \quad \Rightarrow \quad 2\int_0^1 \left(x^{13} - x^{11} + x^9 - x^7 + x^5\right)dx = \frac{11}{24} + a$

$\Rightarrow \quad 2\left(\frac{1}{14} - \frac{1}{12} + \frac{1}{10} - \frac{1}{8} + \frac{1}{6}\right) = \frac{11}{24} + a \quad \Rightarrow \quad a = -\frac{167}{840}$

**5.(C)**




| | |
|---|---|
| D > 0, | f(3) < 0 |
| $(a-1)^2 > 4 \times 2 \times 8$ | $18 - 3a + 3 + 8 < 0$ |
| $a - 1 > 8, a - 1 < -8$ | $3a > 29$ |
| $a < -7, a > 8$ ...(1) | $a > \frac{29}{3}$ ...(2) |

from (1) and (2) $a \in \left(\frac{29}{3}, \infty\right)$

**6. (A)** Operate $R_1 \to R_1 - \sec x R_3$

$$\Rightarrow \quad f(x) = \begin{vmatrix} 0 & 0 & \sec^2 x + \cot x \operatorname{cosec} x - \cos x \\ \cos^2 x & \cos^2 x & \operatorname{cosec}^2 x \\ 1 & \cos^2 x & \cos^2 x \end{vmatrix}$$

$$= \left(\sec^2 x + \cot x \operatorname{cosec} x - \cos x\right)\left(\cos^4 x - \cos^2 x\right)$$

$$= \left(1 + \frac{\cos^3 x}{\sin^2 x} - \cos^3 x\right)\left(\cos^2 x - 1\right)$$

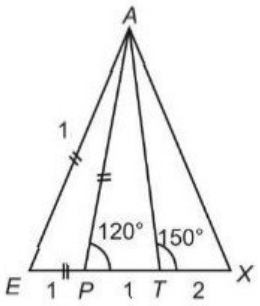
$$= -\sin^2 x \frac{\sin^2 x + \cos^3 x - \cos^3 x \sin^2 x}{\sin^2 x} = -\left(\sin^2 x + \cos^5 x\right)$$

$$\Rightarrow \quad \int_0^{\pi/2} f(x)dx = -\int_0^{\pi/1} \left(\sin^2 x + \cos^5 x\right)dx = -\left(\frac{1}{2}.\frac{\pi}{2} + \frac{4.2}{5.3.1}\right) = -\left(\frac{\pi}{4} + \frac{8}{15}\right)$$

**7.(D)** If $z_1$ be the new complex number then $|z_1| = |z| + \sqrt{2} = 2\sqrt{2}$

Also $\frac{z_1}{z} = \frac{|z_1|}{|z|}e^{i3\pi/2} \Rightarrow z_1 = z.2\left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right) = 2(1+i)(0-i) = -2i + 2 = 2(1-i)$

**8.(D)**




**(A)** $Ax^2 = AE^2 + EX^2 - 2AE.EX\cos\angle AEX = 1+6-2.1.4.\frac{1}{2} = 13 \quad \therefore \quad AX = \sqrt{13}$

**(B)** Since $\angle APT$ isosceles $\angle ATP = \angle PAT = 30^\circ$ then $\angle EAT = 90^\circ$

And also $\frac{AT}{\sin 120^\circ} = \frac{AP}{\sin 30^\circ} \Rightarrow AT = \frac{\sqrt{3}}{2}.\frac{1}{1/2} = \sqrt{3}$

**(C)** Since $EX^2 = AE^2 + AX^2 - 2AE.AX\cos\angle XAE$

$16 = 1+13-2.1\sqrt{13}.\cos\angle XAE$

$\cos\angle XAE = \frac{-1}{\sqrt{13}}$

**9.(B)** $f(x) = ax + b$

$f'(x) = a < 0$ (given) $f(x)$ is decreasing function

So $\begin{matrix} f(-1) = 2(\text{max value}) \Rightarrow & -a+b=2 \\ f(1) = 0 \quad (\text{min value}) & a+b=0 \end{matrix} \Bigg\} \begin{matrix} a=-1 \\ b=1 \end{matrix}$

So $f(x) = -x + 1$

$\Rightarrow \quad f(0) = 1, \; f\left(\frac{1}{4}\right) = \frac{3}{4} \; f(-2) = 3 \; f\left(\frac{1}{3}\right) = \frac{2}{3}$

Now $A = \cos^2\theta + \sin^4\theta$ and we know very well $\frac{3}{4} \le A \le 1$

$\Rightarrow \quad f\left(\frac{1}{4}\right) \le A \le f(0)$

**10.(A)** We have, $\left(\frac{2+\sin x}{y+1}\right)\frac{dy}{dx} = -\cos x, \; y(0) = 1$

$\Rightarrow \quad \int \frac{dy}{y+1} = \int \frac{-\cos x}{2+\sin x} dx$

$\Rightarrow \quad \log|y+1| = -\log|2+\sin x| + c \qquad \ldots\ldots\ldots (i)$

At $x = 0, y = 1$

$\Rightarrow \quad \log|2| = -\log|2+0| + c \quad \Rightarrow \quad c = 2\log 2$

Putting the value of **(c)** the equation (i) becomes

$\log|y+1| = -\log|\sin x + 2| + 2\log 2$

$\Rightarrow \quad (y+1) = \frac{4}{\sin x + 2}$ at $x = \frac{\pi}{2} \Rightarrow y+1 = \frac{4}{\sin\frac{\pi}{2}+2} = \frac{4}{3} \Rightarrow y = \frac{4}{3} - 1 = \frac{1}{3} \Rightarrow y\left(\frac{\pi}{2}\right) = \frac{1}{3}$

**11.(A)** $\dfrac{1}{(1-x)(3-x)}=\dfrac{1}{2}\left[\dfrac{1}{1-x}-\dfrac{1}{3-x}\right]=\dfrac{1}{2}\left[(1-x)^{-1}-(3-x)^{-1}\right]$

$$=\frac{1}{2}\left[(1-x)^{-1}-3^{-1}\left(1-\frac{x}{3}\right)^{-1}\right]=\frac{1}{2}\left[1+x+x^2+x^3+\ldots-\frac{1}{3}\left(1+\frac{x}{3}+\left(\frac{x}{3}\right)^2+\ldots\right)\right]$$

Coefficient of $x^n=\dfrac{1}{2}\left[1-\dfrac{1}{3}.\dfrac{1}{3^n}\right]=\dfrac{3^{n+1}-1}{2.3^{n+1}}$

**12.(A)** Since the system of equations possess non-trivial solution,

$$\begin{vmatrix}\alpha_2 & \alpha_2\\ \beta_1 & \beta_2\end{vmatrix}=0\Rightarrow\alpha_1\beta_2-\alpha_2\beta_1=0$$

$$\Rightarrow\qquad \frac{\alpha_1}{\beta_1}=\frac{\alpha_2}{\beta_2}=\frac{\alpha_1+\alpha_2}{\beta_1+\beta_2}=\left(\frac{\alpha_1\alpha_2}{\beta_1\beta_2}\right)^{1/2}\qquad\ldots\ldots(1)$$

Also $\alpha_1+\alpha_2=-\dfrac{b}{a},\ \alpha_1\alpha_2=\dfrac{c}{a}$ and $\beta_1+\beta_2=-\dfrac{q}{p},\ \beta_1\beta_2=\dfrac{r}{p}$

$$\therefore\qquad \text{Equation (1)}\quad\Rightarrow\quad\frac{-b/a}{-q/p}=\left(\frac{c/a}{r/p}\right)^{1/2}\quad\Rightarrow\quad\frac{b^2}{q^2}=\frac{ac}{pr}.$$

**13.(C)** Coordinates of $F_1$ are $(ae, 0)$ and of $F_2$ are $(-ae, 0)$

Also $BF_1=BF_2=F_1F_2$

$$\Rightarrow\qquad a^2e^2+b^2=4a^2e^2$$

$$\Rightarrow\qquad a^2e^2+a^2\left(1-e^2\right)=4a^2e^2\qquad\Rightarrow\qquad e=\frac{1}{2}\Rightarrow F_1F_2=a$$

And the area of the $\Delta BF_1F_2=\dfrac{\sqrt{3}}{4}a^2$

**14.(B)** We have, $\cot^{-1}\left(\dfrac{n^2-10n+19}{\sqrt{3}}\right)>\dfrac{\pi}{6}$

$$\Rightarrow\qquad\frac{n^2-10n+19}{\sqrt{3}}<\cot\left(\frac{\pi}{6}\right)\qquad\Rightarrow\qquad n^2-10n+25<9$$

$$\Rightarrow\qquad(n-5)^2<3^2\quad\Rightarrow\quad 2<n<8\quad\Rightarrow\quad n\in\{3,4,5,6,7\}$$

Clearly, least value of $n$ is 3

**15.(C)** $\dfrac{dx}{dt}=\cos^2\pi x,$

On differentiating w.r.t. '$x$'

$\dfrac{d^2x}{dt^2}=-2\pi\sin(2\pi x)=$ negative

The particle never reaches point, it means $\dfrac{d^2x}{dt^2}=0\ \Rightarrow\ -2\pi\sin2\pi x=0$

$$\int_0^x\frac{dx}{\sec^2\pi x}=\int_0^t dt$$

$$\frac{\tan\pi x}{\pi}=t\qquad\Rightarrow\qquad\tan(\pi x)=\pi t\ \text{ as }\ t\to\infty\qquad\Rightarrow\qquad x\to\frac{1}{2}$$

**16.(B)** $f(x)=\frac{x^2}{2}+\ln x-2\cos x \quad \Rightarrow \quad f'(x)=x+\frac{1}{x}+2\sin x$

We know that $x+\frac{1}{x}\geq 2,\ \forall\ x>0$ and $2\sin x\leq 2,\ \forall x\in R$

$\Rightarrow \quad f'(x)>0$ for $x>0$

Hence $f(x)$ is increasing in $(0,\infty)$

When $x<0,\ x+\frac{1}{x}\leq -2$ but $-2\leq 2\sin x\leq 2,\ \forall x\in R$

$\Rightarrow \quad f'(0)<0$ hence $f(x)$ is decreasing in $(-\infty, 0)$

**17(C)** Since $g(x)$ is continuous $\forall\ x\in R$, $g(x)$ should be constant

Since $f(x)\in(2,\sqrt{26}),\ a\geq\sqrt{26},\left(as\left[\frac{f(x)}{\sqrt{26}}\right]=0\forall\ x\in R\right)$

So least integral value of $a$ is 6.

**18.(C)** 1S, 3A, 1H, 2R, 1N, 1P, 1U when all letters are different.

Corresponding ways = ${}^7C_3.3!\quad {}^7P_3=210$

When two letters are of one kind and other is different.

Corresponding ways = ${}^2C_1.{}^6C_1.\frac{3!}{2!}=36$

When all letters are alike, corresponding ways = 1.

Thus total words that can be formed = 210 + 36 + 1, i.e. 247

**19.(A)** Total formed numbers that begin with a odd digit = ${}^5C_1.{}^8P_4=5(8)(7)(6)(5)$

Total formed numbers that end with a odd digit = ${}^5C_1.{}^8P_4=(8)(7)(6)(5)$

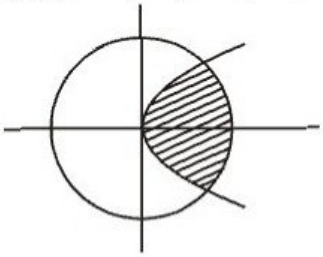
Total formed number that begin with an odd digit and also end with an odd digit

$={}^5C_2.2!.{}^7P_3=5.(4)(7)(6)(5)$

Thus total formed numbers that begin with an odd digit or end with an odd digits is equal to $5\cdot 7\cdot 6\cdot 60$

Total formed numbers = ${}^9P_5=9.8.7.6.5$ Thus, required probability $=\frac{5}{6}$

**20.(A)** The point $(-2k, k+1)$ is the interior point of the circle and parabola



So $(2k)^2+(k+1)^2-4<0 \ \Rightarrow\ 4k^2+k^2+2k+1-4<0 \ \Rightarrow\ 5k^2+2k-3<0$

$(k+1)\left(k-\frac{3}{5}\right)<0\Rightarrow k\in\left(-1,\frac{3}{5}\right)$ ..... (1)

Now $(k+1)^2-4(-2k)<0 \ \Rightarrow\ k^2+2k+1+8k<0 \ \Rightarrow\ k^2+10k+1<0$

$k\in(-5-2\sqrt{6},-5+2\sqrt{6})$ ..... (2)

So from (1) & (2)

$k\in\left(-1,-5+2\sqrt{6}\right)$

## SECTION-2

**1.(0)** Given two lines

$\bar{r} = \bar{a} + t\bar{b}$

$\bar{r} = \bar{c} + s\bar{d}$

$x_2$ is the foot of the perpendicular drawn from $x_1$ on to the second line. Again $x_3$ is the foot of the perpendicular drawn from $x_2$ on to the first line. This process is repeated so that the point of intersection of two lines is obtained.

**2.(11)** $AA^T = I$

$A$ is orthogonal matrix.

$A \times A^T = I$

$$\begin{bmatrix} 2x^2 & 0 & 0 \\ 0 & 6y^2 & 0 \\ 0 & 0 & 3z^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = 2 + 6 + 3 = 11$$

**3.(0)** $\vec{c} = \dfrac{\vec{a} - \mu\vec{b}}{4}$ and $\vec{a}.\vec{b} = 0$

Now, $\left(\vec{b} - \vec{a}\right).\left(\vec{b} + \vec{c}\right) = 0 \Rightarrow \left(\vec{b} - \vec{a}\right).\left(\vec{b} + \dfrac{\vec{a} - \mu\vec{b}}{4}\right) = 0$

$\Rightarrow \quad (4 - \mu)b^2 = a^2 \quad (\because \mu < 4)$ ..... (i)

Again $4|\vec{b} + \vec{c}|^2 = |\vec{b} - \vec{a}|^2$

$\Rightarrow \quad 4\left|\dfrac{(4-\mu)\vec{b} + \vec{a}}{4}\right|^2 = |\vec{b} - \vec{a}|^2 \quad \Rightarrow \quad 4\left(\dfrac{4-\mu}{4}\right)^2 b^2 + \dfrac{a^2}{4} = b^2 + a^2$

$\Rightarrow \left((4-\mu)^2 - 4\right)b^2 = 3a^2$ ..... (ii)

(i) & (ii) we get $\dfrac{(4-\mu)^2 - 4}{4-\mu} = \dfrac{3}{1} \quad \Rightarrow \quad \mu = 0, 5$

$\Rightarrow \quad \mu = 0$ or $5$ but as $\mu < 4$, so, $\mu = 0$.

**4.(14)** $(a+bx)^{-2} = a^{-2}\left(1 + \dfrac{b}{a}x\right)^{-2} = \dfrac{1}{a^2}\left[1 + (-2)\left(\dfrac{b}{a}x\right) + .....\right] = \dfrac{1}{a^2} - \dfrac{2b}{a^3}x + .....$

Also, $(a+bx)^{-2} = \dfrac{1}{4} - 3x + .....$

$\therefore \quad \dfrac{1}{a^2} = \dfrac{1}{4}$ ..... (1)

and $\quad -\dfrac{2b}{a^3} = -3$ ..... (2)

(1) $\Rightarrow \; a^2 = 4 \; \Rightarrow \; a = 2$ and from (2) $b = 12$

**5.(2)** $2\vec{a}-3\vec{b}+6\vec{c}=\vec{0}\Rightarrow 2\vec{a}-3\vec{b}=-6\vec{c}\Rightarrow\left|2\vec{a}-3\vec{b}\right|^2=36|\vec{c}|^2$

$4a^2+9b^2-12\vec{a}.\vec{b}=36c^2$

$4a^2+9b^2-12.ab\cos\theta=36c^2$

$16b^2+9b^2-12.2b^2\cos\theta=36\frac{1}{4}b^2$

$25-24\cos\theta=9\Rightarrow\cos\theta=\frac{2}{3}$

**6.(3)** Side of R are given by

$x=\pm3, y=\pm2$

Let equation of $E_2$ be

$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

As it passes through (0, 4) and (3, 2) we get $\frac{16}{b^2}=1\Rightarrow b^2=16$ and $\frac{9}{a^2}+\frac{4}{b^2}=1=a^2=12$

Eccentricity $e$ of $E_2$ is given by

$a^2=b^2\left(1-e^2\right)$

$\Rightarrow \quad 12=16\left(1-e^2\right)\Rightarrow e=1/2$

**7.(6)** $\sum_{n=1}^{6}\operatorname{cosec}\left(\theta+(n-1)\frac{\pi}{4}\right)\operatorname{cosec}\left(\theta+\frac{n\pi}{4}\right)=\sqrt{2}\left[\cot\theta-\cot\left(\theta+\frac{3\pi}{r}\right)\right]$

$\Rightarrow \quad \cot\theta+\tan\theta=4 \quad\Rightarrow\quad \theta=15^\circ \text{ or } 75^\circ \quad\therefore\quad \sin^2\beta-\sin^2\alpha=\frac{\sqrt{3}}{2}$

**8.(450)** Let the number of $n=x_1, x_2, x_3$.

Since $x_1+x_2+x_3$ is even. That means there are following cases :

**(i)** $x_1, x_2, x_3$ all are even

$\rightarrow$ 4. 5. 5 = 100 ways

**(ii)** $x_1$ even and $x_2, x_3$ are odd

$\rightarrow$ 4.5.5. = 100 ways

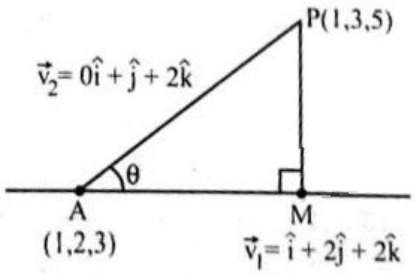
**(iii)** $x_1$ odd, $x_2$ even, $x_3$ odd

$\rightarrow$ 5.5.5 = 125 ways

**(iv)** $x_1$ odd, $x_2$ even, $x_3$ odd

$\rightarrow$ 5.5.5. = 125 ways

**9.(1)** $\therefore \quad PM = |\vec{v}_2|\sin\theta = \sqrt{5}\sin\theta$




$$\text{As, } \cos\theta = \frac{\vec{v}_1.\vec{v}_2}{|\vec{v}_1||\vec{v}_2|} = \frac{6}{3\sqrt{5}} = \frac{2}{\sqrt{5}}$$

$$\Rightarrow \quad \sin\theta = \frac{1}{5}$$

$$\therefore \quad PM = |\vec{v}_2|\sin\theta = \sqrt{5}\left(\frac{1}{\sqrt{5}}\right) = 1$$

**10.(3)** RH Limit $= \lim\limits_{x\to 2}\{[2-(2+h)]+[(2+h)-2]-(2+h)\} = \lim\limits_{h\to 0}\{[-4h]+[h]-2-h\}$

$\lim\limits_{h\to 0}\{0-1-2+h\} = -3$

LH Limit $= \lim\limits_{h\to 0}\{[2-(2-h)]+[(2-h)-2]-(2-h)\} = \lim\limits_{h\to 0}\{0-1-2+h\} = -3$


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